

Instructions:

- **Due February 18, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW03_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework reinforces analytical and computational tools for linear systems, including the matrix exponential solution of state-space models, step and impulse response analysis, and modeling of mechanical systems in state-space form using MATLAB.

Problem 1. Consider the linear system

$$\dot{x} = Ax + Bu. \quad (1)$$

Show that

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau. \quad (2)$$

Hint: Use [Leibniz integral formula](#)

$$\frac{\partial}{\partial t} \int_{a(t)}^{b(t)} f(z, t) dz = f(b(t), t) \frac{\partial b(t)}{\partial t} - f(a(t), t) \frac{\partial a(t)}{\partial t} + \int_{a(t)}^{b(t)} \frac{\partial f(z, t)}{\partial t} dz. \quad (3)$$

Problem 2. Step response. Using the analytical formula (2), solve the scalar differential equation

$$\dot{x}(t) = ax(t) + b,$$

and answer the following.

1. Under what conditions does $\lim_{t \rightarrow \infty} x(t)$ exist? If the limit exists, determine the limiting value. Can you guess the limiting value without solving the equation?
2. Using the analytical solution, write down the solution to the scalar ordinary differential equation

$$\dot{x}(t) = -2x(t) + 8.$$

This equation represents the step response of a linear system, where the constant 8 is the value of the step input.

3. Use Matlab to plot the solutions for the initial conditions $x(0) = 5$ and $x(0) = -4$ in the same figure. Provide a legend to identify each trace, label the axes, and compose a caption for the figure. Determine $\lim_{t \rightarrow \infty} x(t)$ from the plot and compare the numerical value with the analytical limit.
4. Explain how the limiting value of $x(t)$ depends on the constants in the problem, namely, 1) the coefficient of $x(t)$, 2) the height of the step input, and 3) the initial condition.

Problem 3. Consider the following systems

- a) URB, where $m = 1$.
- b) DRB, where $m = c = 1$.
- c) UO, where $m = k = 1$.
- d) DO, where $m = c = 1$ and $k = 2$.
- e) DO, where $m = c = 1$ and $k = 0.25$.
- f) DO, where $m = c = 1$ and $k = 0.2$.

1. In each case, write the system in the state-space form

$$\dot{x} = Ax + Bu, \quad (4)$$

$$y = Cx + Du. \quad (5)$$

Assuming the output is the position, write the A , B , C , and D matrices for each system.

2. Let $x(0) = [1 \ 1]^T$ and $u(t) \equiv 0$. In MATLAB, compute the solution $x(t)$ using the matrix exponential, that is,

$$x(t) = e^{At}x(0). \quad (6)$$

The exponential function e^{At} can be computed using the Matlab function `expm`. Compute the solution from $t = 0$ to $t = 10$ and plot a) $x_1(t)$ against time t , b) $x_2(t)$ against time t , and c) $x_2(t)$ against $x_1(t)$.

3. The impulse response of a linear system is given by $H(t) \triangleq Ce^{At}B$. For each system, calculate and plot the impulse response in Matlab.
4. Use the Matlab function `impz` to compute the impulse response in each of the cases above, and plot them on the same figure. Do they match up?